

ShopFlow AWS Project

CI/CD Pipeline Design

CodeBuild - CodePipeline - Source to Build to Deploy

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Field	Value
Version	1.0
Date	June 2026
Region	ap-south-1 (Mumbai)
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Application	ShopVerse -- Spring Boot 3.2.5 / Java 21

1. Overview

ShopFlow uses AWS CodeBuild and CodePipeline to automate the full CI/CD workflow. Every push to the main branch in GitHub triggers an automated pipeline: Source -- Build -- Deploy. Total deploy time is approximately 10-15 minutes.

2. Pipeline Stages

Stage	Service	Input	Output	Duration
Source	CodePipeline (GitHub)	Git push to main	Source code ZIP in S3	~10s
Build	CodeBuild	Source code ZIP from S3	imagedefinitions.json in S3	~5-8 min
Deploy	ECS rolling update	imagedefinitions.json	New ECS task running	~3-5 min

3. CodeBuild Project -- shopflow-build-dev

Setting	Value	Notes
Build Image	aws/codebuild/standard:7.0	Amazon Linux 2023 + Java 21 + Docker
Compute	BUILD_GENERAL1_SMALL	3 GB RAM, 2 vCPU -- free tier (100 min/month)
Privileged Mode	Enabled	Required for Docker build inside CodeBuild
Maven Cache	S3 /cache folder	Caches .m2 dependencies between builds
Service Role	shopflow-cicd-role	Push to ECR, write to S3 artifacts bucket

4. buildspec.yml Phases

Phase	Commands
install	runtime-versions: java: corretto21
pre_build	Authenticate to ECR: aws ecr get-login-password docker login Set IMAGE_TAG = first 8 chars of git commit hash
build	./mvnw clean package -DskipTests -pl shopverse-web -am docker build -t \$ECR_REPO_URI:\$IMAGE_TAG .
post_build	docker push \$ECR_REPO_URI:\$IMAGE_TAG docker push \$ECR_REPO_URI:latest Write imagedefinitions.json

5. imagedefinitions.json

The build output that tells CodePipeline's ECS deploy action which image to use:

```
[{"name":"shopverse","imageUri":"123456789.dkr.ecr.ap-south-1.amazonaws.com/shopflow-shopverse:a1b2c3d4"}]
```

6. GitHub Connection -- One-Time Manual Setup

Step	Action
1	AWS Console -- CodePipeline -- Settings -- Connections -- Create connection
2	Select GitHub -- Name it 'shopflow-github'
3	Click 'Connect to GitHub' -- Authorize in GitHub
4	Copy the connection ARN -- paste it into CloudFormation parameter GitHubConnectionArn
5	Connection status must be AVAILABLE before deploying the pipeline stack

7. Rollback Strategy

Method	How	Time to Recover
ECS circuit breaker (automatic)	Health check fails 3x -- ECS reverts to previous task definition	~3 minutes
Manual re-run pipeline	Update ImageUri parameter to previous tag -- re-run stack update	~5 minutes
Force ECS task definition	aws ecs update-service with previous task definition revision	~2 minutes